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[AI-BASED GAP ANALYSIS OF ERP SYSTEM USAGE IN RETAIL ORGANISATIONS IN BAHRAIN]

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ABSTRACT

This study investigates the gap between expected and actual ERP system usage in the Bahraini retail sector using Machine Learning techniques. The research combined a case study dataset focused on continuance intention with a survey dataset representing actual ERP usage within organizations. Data were analyzed using WEKA under the CRISP-DM framework through J48, Random Forest, and Naïve Bayes algorithms.

The findings revealed that organizational and human factors such as monitoring, training, ICT skills, management support, and information quality significantly influence ERP system utilization. The results also showed that many employees rely on external tools instead of fully utilizing ERP system functions, which reduces the effectiveness of organizational operations and decision-making processes.

Among the tested models, Random Forest achieved the highest accuracy in the case study dataset, while J48 produced the best results in the survey scaled model due to its ability to capture direct organizational decision paths. The study confirmed the existence of a clear gap between users' intention and actual ERP usage in practice. Overall, the research demonstrates how Machine Learning techniques can help organizations identify ERP usage gaps and support more effective ERP adoption strategies in retail environments.

OBJECTIVES

This study aims to investigate ERP underutilisation in Bahraini retail organisations and examine how AI/ML techniques can help identify ERP usage gaps and improve organisational performance. However, many organisations still face challenges related to ERP underutilisation, where several system functions and modules are not fully used in practice. This study focuses on identifying these usage gaps within the Bahraini retail sector by applying AI/ML-based analytical techniques.

- To identify key indicators of ERP underutilisation in retail organisations in Bahrain.
- To develop and apply an AI/ML-based analytical model for detecting ERP usage gaps using WEKA.
- To evaluate the performance of the developed models and examine the relationship between ERP utilisation and organisational performance.
- To analyse the organisational and behavioural factors influencing ERP system usage in the Bahraini retail sector.
- To provide practical insights that support organisations in improving ERP adoption and operational efficiency.

Overall, these objectives contribute to developing a structured and data-driven approach for understanding ERP usage behaviour and supporting more effective ERP utilisation within retail organisations in Bahrain. This study also aims to provide both academic and practical insights for improving ERP effectiveness in the Bahraini retail sector.

METHODS/DIAGRAMS/FIGURES

Aspect	Case Study (South Korea)	Survey (General Model)	Survey (Scaled Model)
Target Variable	Continuance Intention	Actual ERP Usage Gap	ERP Gap (Construct-based)
Best Model	Random Forest (~89%)	J48 (~71.6%)	J48 (~75.7%)
Nature of Data	Behavioural/perceptual	Organisational + behavioural	Theoretical constructs
Dominant Factors	Usefulness, system quality, satisfaction	Monitoring, training, and external tools	Information quality, skills, and top management
Framework Alignment	DeLone & McLean	TOE (operational level)	TOE + IS Success (integrated view)
Key Insight	Users are willing to use ERP	ERP is not always used effectively	Root causes of the usage gap

Table 13: Comparison Between Case Study and Survey

Figure 24: J48 decision tree of the survey generated by Weka (Survey)

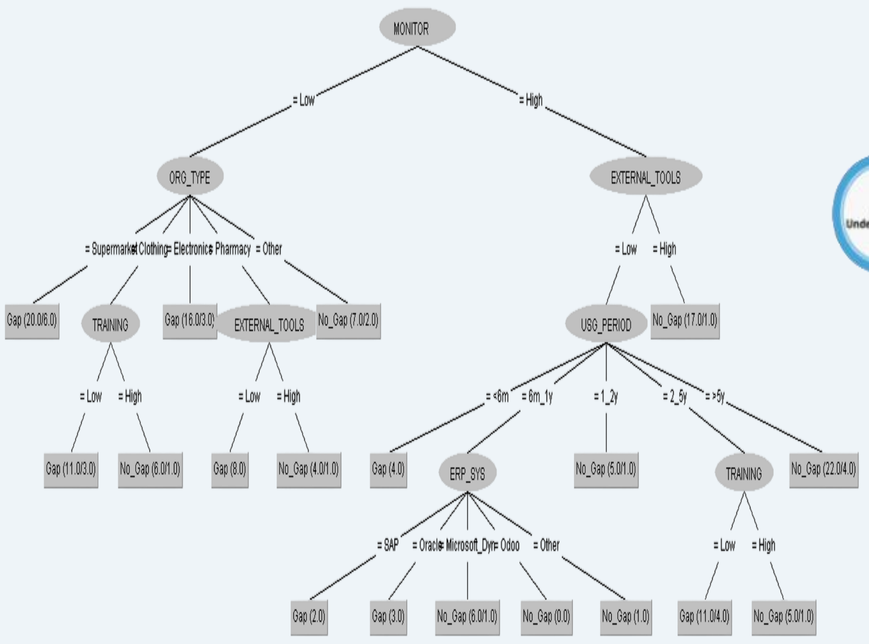
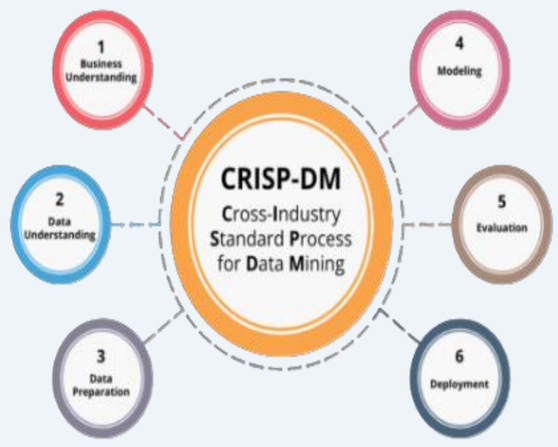


Figure 19: CRISP-DM Framework



	PREDICTED: LOW (A)	PREDICTED: HIGH (B)
ACTUAL: LOW (A)	51	26
ACTUAL: HIGH (B)	16	55

Table 8: Confusion matrix of J48 algorithm (Survey)

RESULTS

The analysis revealed a noticeable gap between users' intention to use ERP systems and their actual usage in retail organizations. Machine Learning models successfully identified the key organizational and behavioral factors influencing ERP underutilization. The findings also showed that ERP implementation success depends not only on system quality, but also on employee readiness, management support, and effective organizational practices.

- Random Forest achieved the highest accuracy in the case study dataset (~89%).
- J48 produced the best results in the survey scaled model (~75.7%).
- Many employees depended on external tools instead of fully using ERP systems.
- Monitoring, training, ICT skills, and information quality were major factors affecting ERP usage.
- The findings confirmed that organizational conditions strongly influence effective ERP adoption.

CONCLUSION & FUTURE WORK

The study concluded that ERP usage gaps in Bahraini retail organizations are mainly caused by organizational and human-related factors rather than technical limitations. Although users showed positive intention toward ERP systems, this intention did not always translate into effective real-world usage. The findings also demonstrated that Machine Learning techniques can successfully identify the factors contributing to ERP underutilization and support better organizational decision-making.

FUTURE WORK:

- Develop real-time ERP monitoring dashboards.
- Expand the study to sectors beyond retail.
- Integrate process mining with Machine Learning techniques.
- Increase dataset size for stronger generalization.
- Conduct long-term studies on ERP usage trends.